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N-body Simulations of Spiral Bending Waves in Planetary Rings

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ABSTRACT

Draft. This mock paper currently holds the derivation of the vertical epicyclic motion used by the epi_int_3d drift step (Section 2), the vertical gravity that one streamline exerts on another (Section 3), its relation to epi_int_lite's in-plane streamline gravity (Appendix B), and a study of the streamline method's resolution limit (Appendix C).

1. INTRODUCTION

The epi_int and epi_int_lite N-body integrators (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025) model a planetary ring as a set of gravitating streamlines whose particles are confined to the ring plane. Spiral bending waves, which are launched at a satellite's vertical resonances (F. H. Shu et al. 1983; F. H. Shu 1984), require the particles' vertical motions too. This paper describes epi_int_3d, a modest extension of epi_int_lite into the vertical (out-of-plane) direction, and uses it to simulate spiral bending waves.

2. VERTICAL EPICYCLIC MOTION

The epi_int_lite drift step advances each particle's unperturbed orbit about an oblate planet using first-order epicyclic equations that are exactly reversible: converting spatial coordinates into epicyclic orbit elements and back introduces no error, so the drift step is not a source of numerical drift (J. M. Hahn et al. 2025). Here those equations are extended to particles having vertical displacement z and vertical velocity v_z, with the same first-order accuracy and the same exact reversibility.

2.1. *Equations of motion*

A particle orbiting an oblate planet obeys Newton's second law,

$$\ddot{\mathbf{r}} = -\nabla\Phi, \tag{1}$$

where Φ is the planet's gravitational potential. That potential is a series in the planet's zonal harmonics J_{2k} (Eqn. 5 of N. Borderies-Rappaport & P.-Y. Longaretti 1994), and as in J. M. Hahn & J. N. Spitale (2013) and J. M. Hahn et al. (2025) only the J_2 term is kept, so

$$\Phi(r,z) = -\frac{GM_p}{\rho}\left[1 - J_2\left(\frac{R_p}{\rho}\right)^2 P_2\left(\frac{z}{\rho}\right)\right]. \tag{2}$$

Here M_p is the planet's mass, R_p its radius, r the particle's cylindrical radius, $\rho^2 = r^2 + z^2$, and $P_2(\mu) = (3\mu^2 - 1)/2$. In cylindrical coordinates (r, θ, z), Eqn. (1) is

$$\begin{aligned} \ddot{r} - r\dot{\theta}^2 &= -\frac{\partial\Phi}{\partial r} \\ \frac{d}{dt}\left(r^2\dot{\theta}\right) &= -\frac{\partial\Phi}{\partial\theta} = 0 \\ \ddot{z} &= -\frac{\partial\Phi}{\partial z}. \end{aligned} \tag{3}$$

The potential is independent of θ, so the z-component of the specific angular momentum, $h = r^2\dot{\theta} = rv_\theta$, is conserved exactly, even when $z \neq 0$. The potential is also even in z, so near the ring plane

$$\Phi(r,z) = \Phi(r,0) + \frac{1}{2}\nu^2(r)z^2 + \mathcal{O}(z^4), \tag{4}$$

where $\Phi(r,0) = -GM_p/r - J_2GM_pR_p^2/2r^3$ is the midplane potential and

$$\nu^2(r) = \left.\frac{\partial^2\Phi}{\partial z^2}\right|_{z=0} = \frac{GM_p}{r^3}\left[1 + \frac{9}{2}J_2\left(\frac{R_p}{r}\right)^2\right]. \tag{5}$$

Equations (3) then become

$$\begin{aligned} \ddot{r} &= \frac{h^2}{r^3} - \frac{\partial\Phi(r,0)}{\partial r} + \mathcal{O}(z^2) \\ \ddot{z} &= -\nu^2(r)z + \mathcal{O}(z^3). \end{aligned} \tag{6}$$

To first order the radial equation does not involve z; it is the planar equation of motion. J. M. Hahn et al.

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(2025) solved the radial part of Eqns. (6) to first order in small displacements from circular motion, in a form that is exactly reversible: the conversion from the coordinates r, θ, v_r, v_θ to the epicyclic elements $a, e, \tilde{\omega}, M$, and back, is exact. This is required if the solution is to be used in the drift step of a drift-kick N-body integrator. That solution is summarized in Appendix A. In it the particle's radius is $r = a[1 + \mathcal{O}(e)]$, its mean longitude $\lambda = \tilde{\omega} + (\Omega_a/\kappa_a)M$ advances at the mean angular velocity Ω_a, and its mean anomaly M advances at the epicyclic frequency κ_a. Note that Ω denotes the longitude of ascending node, while Ω_a is the mean angular velocity.

So the only additional physics that epi_int_lite's drift step needs in order to follow out-of-plane motion is the vertical equation of motion, the second of Eqns. (6); the vertical forces that the ring and satellites exert enter the kick step and are treated separately. Evaluating $\nu^2(r)$ at $r = a$ reduces the vertical equation to the harmonic oscillator

$$\ddot{z} = -\nu_a^2 z, \tag{7}$$

with vertical frequency (Eqn. 17 of N. Borderies-Rappaport & P.-Y. Longaretti 1994)

$$\nu_a \equiv \nu(a) = \sqrt{\frac{GM_p}{a^3}}\left[1 + \frac{9}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}, \tag{8}$$

where the neglected terms are of order $\mathcal{O}(ei)$, due to the particle's radial excursion $r - a = \mathcal{O}(ea)$, and $\mathcal{O}(i^3)$. Outside the planet the potential obeys Laplace's equation, which for an axisymmetric potential is

$$\nabla^2\Phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\Phi}{\partial r}\right) + \frac{\partial^2\Phi}{\partial z^2} = 0. \tag{9}$$

At $r = a$ and $z = 0$, where $\Omega_a^2 = r^{-1}\partial\Phi/\partial r$, $\kappa_a^2 = 3\Omega_a^2 + \partial^2\Phi/\partial r^2$ and $\nu_a^2 = \partial^2\Phi/\partial z^2$, Eqn. (9) relates the three frequencies via $\kappa_a^2 + \nu_a^2 = 2\Omega_a^2$.

2.2. *Vertical solution*

The solution of Eqn. (7) is a sinusoid of frequency ν_a whose amplitude and phase define two new orbit elements, the inclination i and the longitude of ascending node Ω:

$$\begin{aligned} z &= a\, i \sin(\lambda - \Omega) \\ v_z &= a\nu_a i \cos(\lambda - \Omega), \end{aligned} \tag{10}$$

where the node regresses at the rate

$$\dot{\Omega} = \Omega_a - \nu_a \simeq -\frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\Omega_a. \tag{11}$$

The vertical phase $\lambda - \Omega$ then advances at exactly ν_a, so Eqns. (10) solve Eqn. (7) exactly, with $v_z = \dot{z}$, and the particle crosses the ring plane upward at longitude $\lambda = \Omega$. Measuring the phase from λ rather than θ is what keeps the vertical motion a pure harmonic oscillator; $\theta - \Omega$ would add an $\mathcal{O}(ei)$ error to the phase.

2.3. *Coordinates to elements*

Equations (10) are inverted by a fifth step that follows steps 1–4 of Appendix A. Step 1 remains exact in three dimensions because h is conserved and r is the cylindrical radius, so steps 1–4 provide a, Ω_a, κ_a, e, M, $\tilde{\omega}$, and so λ. Step 5 computes ν_a from Eqn. (8) and then

$$\begin{aligned} i &= \sqrt{\left(\frac{z}{a}\right)^2 + \left(\frac{v_z}{a\nu_a}\right)^2} \\ \Omega &= \lambda - \mathrm{atan2}(z, v_z/\nu_a). \end{aligned} \tag{12}$$

Once a is known, the mapping $(i, \lambda - \Omega) \leftrightarrow (z, v_z/\nu_a)$ is a change between polar and Cartesian coordinates, so Eqns. (10) and (12) are exact inverses. The drift step from t to $t + \Delta t$ converts coordinates to elements, advances $M \to M + \kappa_a\Delta t$ and $\Omega \to \Omega + (\Omega_a - \nu_a)\Delta t$, and converts back. Its vertical part is simply a rotation of $(z, v_z/\nu_a)$ through the angle $\nu_a\Delta t$.

3. THE GRAVITY OF A STREAMLINE

Epi_int and epi_int_lite assume that the gravitational acceleration of a particle by perturbing streamline a' is that due to a straight and infinitely long wire:

$$\mathbf{A}_g = \frac{2G\lambda'\boldsymbol{\Delta}}{\Delta^2}, \tag{13}$$

where λ' is the perturbing streamline's linear density and $\boldsymbol{\Delta}$ is the separation of the particle to the streamline's nearest point, designated the near point in Fig. 1. The following derives $\boldsymbol{\Delta}$ in three dimensions, and Appendix B relates that to the co-planar problem considered by epi_int and epi_int_lite.

3.1. *The perturbing streamline's geometry*

All epi_int_3d simulations span only a narrow annulus within a broader planetary ring, so every particle is radially close to the nearby portions of all perturbing streamlines. Which allows us to treat the particle's acceleration due to a perturbing streamline as that due to a straight wire, and the following 3D treatment is an extension of the 2D force calculations used in J. M. Hahn & J. N. Spitale (2013).

In cylindrical coordinates the particle is at $\mathbf{r} = (r, \theta, z)$, while the perturbing streamline's coordinates $\mathbf{r}' = (r', \theta', z')$ are to be regarded as functions of its longitude $\theta' = \theta + \delta\theta'$ relative to the particle's longitude θ. Streamline a' is sampled by its particles, so Lagrange

polynomials are fit to the three particles in a' that are nearest in longitude, which allows $r'(\theta')$, $z'(\theta')$ and the linear density $\lambda'(\theta')$ to be regarded as smooth functions of θ'. And because the particle is close to the streamline, $|r'-r| \ll r$, $|z'-z| \ll r$ and $|\delta\theta'| \ll 1$, so the particle's local frame $(\hat{r}, \hat{\theta}, \hat{z})$ can be treated as Cartesian.

The vector from the particle to the streamline's near point (the black square in Fig. 1) is then

$$\mathbf{\Delta} = \mathbf{r}'(\theta + \delta\theta') - \mathbf{r} = \mathbf{d}_0 + \mathbf{d}_1, \tag{14}$$

where $\delta\theta'$ is the near point's longitude relative to the particle's longitude. This splits the trip to the near point across two legs. The first leg, $\mathbf{d}_0 = \mathbf{r}'(\theta) - \mathbf{r}$, travels from the particle to the crossing point $\mathbf{r}'(\theta) = (r'(\theta), \theta, z'(\theta))$, which is where the streamline crosses the particle's longitude (the cross in Fig. 1), so $\mathbf{d}_0 = (d_r, 0, d_z)$ in the local frame, where $d_r = r'(\theta) - r$ and $d_z = z'(\theta) - z$. The second leg, $\mathbf{d}_1 = \mathbf{r}'(\theta + \delta\theta') - \mathbf{r}'(\theta)$, then runs along the streamline from the crossing point to the near point. That stretch of streamline is short and locally straight, so $\mathbf{d}_1 \simeq [r'(\theta + \delta\theta') - r'(\theta), r\,\delta\theta', z'(\theta + \delta\theta') - z'(\theta)] \simeq (s_r, 1, s_z)\, r\,\delta\theta' = Nr\,\delta\theta'\,\hat{\mathbf{t}}$ to first order in $\delta\theta'$, where the streamline's unit tangent is

$$\hat{\mathbf{t}} = \frac{(s_r, 1, s_z)}{N}, \qquad N^2 = 1 + s_r^2 + s_z^2, \tag{15}$$

and its slopes are

$$s_r = \frac{1}{r}\frac{dr'}{d\theta'}, \qquad s_z = \frac{1}{r}\frac{dz'}{d\theta'} \tag{16}$$

at $\theta' = \theta$, which are taken from the derivatives of the Lagrange polynomials.

The near point is where $\mathbf{\Delta}$ is perpendicular to the streamline, $\mathbf{\Delta} \cdot \hat{\mathbf{t}} = 0$. With $\mathbf{d}_1 = Nr\,\delta\theta'\,\hat{\mathbf{t}}$, that condition becomes $\mathbf{d}_0 \cdot \hat{\mathbf{t}} + Nr\,\delta\theta' = 0$, so the near point lies at $\delta\theta' = -c/r$, where $c = (\mathbf{d}_0 \cdot \hat{\mathbf{t}})/N = (d_r s_r + d_z s_z)/N^2$, and

$$\mathbf{\Delta} = \mathbf{d}_0 - (\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}} = \hat{\mathbf{t}} \times (\mathbf{d}_0 \times \hat{\mathbf{t}}), \tag{17}$$

which is simply $\mathbf{d}_0$ with its component along the streamline removed. In components,

$$\begin{aligned} \mathbf{\Delta} &= (d_r - c s_r, -c, d_z - c s_z) \\ \Delta^2 &= d_r^2 + d_z^2 - c(d_r s_r + d_z s_z). \end{aligned} \tag{18}$$

Note that $|c| \le |d_r s_r| + |d_z s_z| \ll r$, so $|\delta\theta'| \ll 1$ as assumed, and the near point lies much closer to the crossing point than the spacing of the streamline's particles. Inserting $\mathbf{\Delta}$ into Equation (13) then provides the acceleration that streamline a' exerts on the particle.

In the ring plane $d_z = s_z = 0$, and the above reduces to the geometry of Fig. 3 of J. M. Hahn & J. N. Spitale (2013), which is redrawn here as Fig. 1. Let φ be the angle at which the streamline crosses $\hat{\theta}$, so that $s_r = \tan\varphi$. Then $c = d_r \sin\varphi\cos\varphi$ and $\mathbf{\Delta} = d_r \cos\varphi(\cos\varphi, -\sin\varphi, 0)$, i.e., the perpendicular has length $\Delta = d_r\cos\varphi$ and makes the angle φ with $\hat{r}$. The acceleration is then

$$\begin{aligned} A_{g,r} &= \frac{2G\lambda'}{d_r} \\ A_{g,\theta} &= -\frac{2G\lambda'\tan\varphi}{d_r}. \end{aligned} \tag{19}$$

Out of the plane, the tangential acceleration $A_{g,\theta} = -2G\lambda' c/\Delta^2$ also includes the term $-2G\lambda' d_z s_z/N^2\Delta^2$, which is due to the streamline's vertical tilt, and this is how vertically displaced streamlines exchange angular momentum.

Equations (13) and (14–19) are exact for a straight wire. But the streamline is curved, with a radius of curvature $R \simeq r$, and that curvature introduces two errors. First, the streamline bends away from its tangent between the crossing point and the near point, which rotates $\mathbf{\Delta}$ by the angle $(\mathbf{d}_0 \cdot \hat{\mathbf{t}})/R$ and changes its length by the fraction $(\mathbf{d}_0 \cdot \hat{\mathbf{t}})^2/2R\Delta$. Relative to $A_{g,\theta}$ and $A_{g,r}$, these errors are of order Δ/R and $s_r^2\Delta/R$. Second, a curved streamline does not pull exactly like a straight wire. For instance, a circular ring of matter exerts the acceleration $(2G\lambda'/\Delta)[1 \pm (\Delta/2R)(\ln(8R/\Delta) - 2)]$ on a point in its plane a distance Δ away, with the sign $+$ outside the ring and $-$ inside. This is the error of the wire approximation itself (J. M. Hahn & J. N. Spitale 2013), which is of the same order as the first error: about 5×10^{-4} for a streamline $\Delta = 10^{-4}R$ away, and smaller still for nearer streamlines.

3.2. *The restoring force of a bending wave*

Equation (13) recovers the self-gravity of a bending wave. Consider a sheet of surface density σ made of streamlines with $\lambda = \sigma\,dr'$, displaced vertically to $Z(r) = Z_0 \cos kr$. When $|d_z| \ll |d_r|$ and the slopes are small, $A_{g,z} \simeq 2G\lambda d_z/d_r^2$, and the sum of $A_{g,z}$ over the sheet is

$$A_z = 2G\sigma Z_0 \int_{-\infty}^{\infty} \frac{\cos kx - 1}{x^2}\,dx = -2\pi G\sigma|k|Z, \tag{20}$$

which is the self-gravity term of the bending-wave dispersion relation, $(\omega - m\Omega_a)^2 = \nu_a^2 + 2\pi G\sigma|k|$ (F. H. Shu et al. 1983; F. H. Shu 1984). So summing the wires' pulls over the streamlines gives the vertical restoring force of a self-gravitating sheet, just as the same sum gives the $-2\pi G\sigma|k|$ term of a density wave in the plane. For bending waves, though, self-gravity is restoring, and adds to ν_a^2.

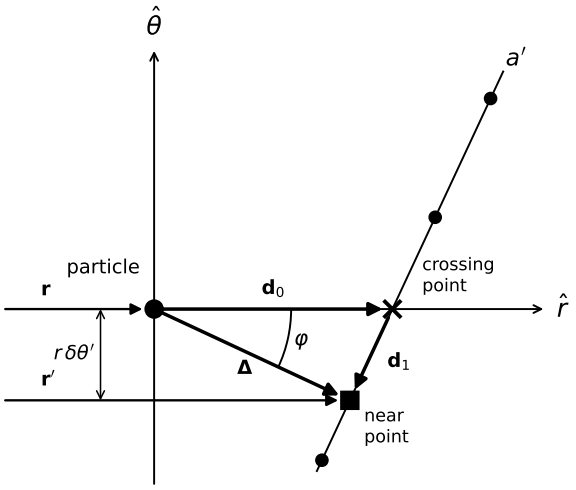


Figure 1. The gravity of a perturbing streamline a' on a particle in the ring plane: Fig. 3 of J. M. Hahn & J. N. Spitale (2013), redrawn. Near the particle's longitude, a' (thin line, with the three particles nearest in longitude) is straight, tilted from $\hat{\theta}$ by φ, where $\tan\varphi = s_r$, and it crosses that longitude at $\mathbf{d}_0 = d_r\hat{r}$ from the particle (cross). Its near point (black square) lies $\mathbf{d}_1$ further along it, $r\,\delta\theta'$ along $\hat{\theta}$ from the crossing point (here $\delta\theta' < 0$). The perpendicular $\boldsymbol{\Delta} = \mathbf{r}' - \mathbf{r} = \mathbf{d}_0 + \mathbf{d}_1$ (Eqn. 14), where $\mathbf{r}$ and $\mathbf{r}'$ are the positions of the particle and the near point (the planet is far off to the left), has length $\Delta = d_r\cos\varphi$ and makes the angle φ with $\hat{r}$. The wire's acceleration $A_g = 2G\lambda'/\Delta$ points along it (Eqns. 19).

On the code's lattice of streamlines spaced Δ apart (here Δ is the spacing, as in Appendix C), with $Z_j = Z_0\cos(j\phi)$ and $\phi = k\Delta$,

$$A_z = \frac{4G\sigma Z_0}{\Delta}\sum_{n=1}^{\infty}\frac{\cos n\phi - 1}{n^2} = -X(\phi)Z_0, \tag{21}$$

where $X(\phi)$ is the lattice self-gravity of Eqn. (C4). The vertical lattice modes therefore oscillate at $\omega^2 = \nu_a^2 + X(\phi)$, which tends to the continuum value as $\phi \to 0$. At the zigzag $\phi = \pi$, the extra restoring force is half the continuum value, as it is in the plane. Vertically, however, the lattice's self-gravity is stabilizing, so the resolution limit of Eqn. (C7), which concerns in-plane modes, has no vertical counterpart.

3.3. *Implementation*

Epi_int_3d computes Eqn. (13), with λ' the perturbing streamline's own linear density at the particle's longitude, and both slopes s_r and s_z taken from the derivatives of the Lagrange polynomials that give r' and z' there. In the plane it is Eqns. (19). For regression tests, epi_int_3d can instead use epi_int_lite's ring gravity (Eqns. B3), with which it reproduces epi_int_lite's results exactly. Δ never vanishes, since streamlines may not cross. As in epi_int_lite, the particle's own streamline exerts no force on it. The slopes require that the streamline be well sampled by its particles.

APPENDIX

A. THE IN-PLANE SOLUTION OF HAHN ET AL. (2025)

This appendix summarizes the first-order, exactly reversible solution of the planar equation of motion from Appendix A of J. M. Hahn et al. (2025), in the notation used here. The mean angular velocity and epicyclic frequency at semimajor axis a are

$$\Omega_a = \sqrt{\frac{GM_p}{a^3}}\left[1 + \frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}$$
$$\kappa_a = \sqrt{\frac{GM_p}{a^3}}\left[1 - \frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}. \tag{A1}$$

In terms of the epicyclic elements a, e, $\tilde{\omega}$ and M, the particle's coordinates are

$$r = \frac{a}{\sqrt{1 + 2e\cos M}}$$
$$\theta = \frac{\Omega_a}{\kappa_a}(M + 2e\sin M) + \tilde{\omega}$$
$$v_r = ea\kappa_a\sin M\left(\frac{r}{a}\right)^3$$
$$v_\theta = \frac{a^2\Omega_a}{r}, \tag{A2}$$

which conserve $h = rv_\theta = a^2\Omega_a$ exactly and solve the planar equation of motion with errors of order $\mathcal{O}(e^2)$. The elements advance as $M \to M + \kappa_a \Delta t$, with a, e and $\tilde{\omega}$ constant. Equations (A2) are inverted exactly by:

1. Compute $h = rv_\theta$, then solve $h^2 = a^4\Omega_a^2$ for the semimajor axis, $a = R_p\left(C + \sqrt{C^2 - 3J_2/2}\right)$ where $C = h^2/2GM_pR_p$.
2. Compute Ω_a and κ_a from Eqns. (A1).
3. Compute $e\sin M = (v_r/a\kappa_a)(a/r)^3$ and $e\cos M = \frac{1}{2}[(a/r)^2 - 1]$, which give e and M.
4. Compute $\tilde{\omega} = \theta - (\Omega_a/\kappa_a)(M + 2e\sin M)$.

B. RELATION TO EPI_INT AND EPI_INT_LITE

In the plane, the pull of Section 3.1 reduces to Eqns. (19), the pull of Fig. 3 of J. M. Hahn & J. N. Spitale (2013) (Fig. 1). J. M. Hahn & J. N. Spitale (2013) give its components as $A_{g,r} = A_g\cos\varphi$ and $A_{g,\theta} = -A_g\sin\varphi$, with $A_g = 2G\lambda/\Delta$ and Δ the perpendicular distance, and approximate them as $A_{g,r} \simeq A_g$ and $A_{g,\theta} \simeq -A_g v_r'/v_\theta'$ (their Eqns. 20), to lowest order in the streamline's eccentricity. With Δ read as the radial gap d_r and $\tan\varphi = v_r'/v_\theta'$, those are Eqns. (19) without approximation. Epi_int_lite (J. M. Hahn et al. 2025) differs from Eqns. (19) in three ways, which epi_int_3d corrects.

Distance—Epi_int_lite evaluates the pull's magnitude at the crossing point, $\Delta = d_r$, but keeps its direction toward the near point,

$$A_{g,r} = \frac{2G\lambda\cos\varphi}{d_r}$$
$$A_{g,\theta} = -\frac{2G\lambda\sin\varphi}{d_r}. \tag{B3}$$

This pull is weaker than Eqns. (19) by the factor $\cos\varphi = 1 - \mathcal{O}(\varphi^2)$. In a planetary ring φ is of order the streamlines' eccentricities, so the difference is $\sim 10^{-10}$ and of no practical consequence, but Eqns. (19) cost no more.

Direction—Epi_int_lite takes the wire's direction from the streamline's velocity at the crossing point, $\sin\varphi = v_r'/v'$ and $\cos\varphi = v_\theta'/v'$ (Fig. 2). The velocity is tangent to the streamline only when the streamline's shape is stationary. If the shape is a pattern that rotates at Ω_p while its particles orbit at Ω, then $v_r'/v_\theta' = (1 - \Omega_p/\Omega)s_r$, where s_r is the slope of the streamline's shape (Eqn. 16). Near an m-armed Lindblad resonance $1 - \Omega_p/\Omega \simeq \pm 1/m$, so the velocity underestimates the wire's tilt, and with it the tangential pull that transports angular momentum between streamlines, by a factor of about m. For the slowly precessing $m = 1$ ringlets of J. M. Hahn et al. (2025), Ω_p is the precession rate and the error is negligible. Epi_int_3d takes the direction from the shape (Eqn. 15).

Linear density—Epi_int_lite evaluates every perturbing streamline's pull with the perturbed particle's own linear density λ, rather than the perturbing streamline's λ'. The difference, $\lambda' - \lambda \simeq (a' - a)\partial\lambda/\partial a$, adds a nearly uniform pull of order the ring's mass over the planet's; in the eccentric ringlets of J. M. Hahn et al. (2025) it changes the precession rate by about 10^{-7} of itself.

In an eccentric ringlet the three corrections together change the streamlines' eccentricities by less than 10^{-4} of themselves over 5000 orbits. Only the second matters for spiral waves, where it changes the torque between streamlines by a factor of about m. Epi_int_3d keeps epi_int_lite's ring gravity, Eqns. (B3), as an option, so that its results can be checked against epi_int_lite's exactly.

C. RESOLUTION LIMIT OF THE STREAMLINE METHOD

A streamline model represents a continuous ring by N_r streamlines spaced Δ apart, and that discreteness sets which wavelengths of the ring's response it can follow. This appendix derives the axisymmetric stability of a uniform lattice of circular streamlines under the self-gravity and pressure accelerations of epi_int_lite (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025), and compares it with the continuum Toomre criterion.

Each streamline of linear density λ exerts the acceleration $2G\lambda/d$ of a straight wire on a particle a distance d away (J. M. Hahn & J. N. Spitale 2013). The pressure $p = c^2\sigma$ acts through the surface density $\sigma_j = 2\lambda/(r_{j+1} - r_{j-1})$ of

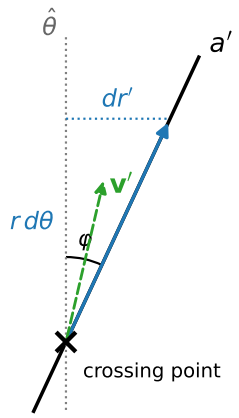


Figure 2. The perturbing streamline at its crossing point (Fig. 1). Its shape has the slope $\tan\varphi = s_r$: a step $r\,d\theta$ along $\hat{\theta}$ moves it dr' outward. Its velocity $\mathbf{v}'$ (dashed), which epi_int_lite takes as the wire's direction, is tilted less when the shape rotates, here by half as much, as for an $m = 2$ pattern.

streamline j and the central difference $A_j = -(p_{j+1} - p_{j-1})/2\lambda$, which tends to $-\sigma^{-1}\partial p/\partial r$ as $\Delta \to 0$. Displace the streamlines radially by $\xi_j = \xi\cos(j\phi)$, where $\phi = k\Delta$ and k is the radial wavenumber, so that $0 < \phi \le \pi$; $\phi = \pi$ is the shortest wave the lattice can carry, the zigzag $\xi_j = (-1)^j\xi$ of Fig. 3a. Linearizing in ξ, the streamlines at $j = \pm n$ add $4G\lambda(1 - \cos n\phi)\xi/(n\Delta)^2$ to the acceleration of streamline 0, and summing over n with $\sigma = \lambda/\Delta$ gives

$$X(\phi) = \frac{4G\sigma}{\Delta}\sum_{n=1}^{\infty}\frac{1-\cos n\phi}{n^2} = \frac{4G\sigma}{\Delta}\left(\frac{\pi\phi}{2} - \frac{\phi^2}{4}\right), \tag{C4}$$

while pressure contributes

$$\omega_p^2(\phi) = \left(\frac{c}{\Delta}\right)^2\sin^2\phi, \tag{C5}$$

so that each mode oscillates at the frequency

$$\omega^2 = \kappa^2 - X(\phi) + \omega_p^2(\phi). \tag{C6}$$

For $\phi \ll 1$ these recover the continuum terms $X \to 2\pi G\sigma k$ and $\omega_p^2 \to c^2k^2$, and Eqn. (C6) becomes the dispersion relation of a pressure-supported self-gravitating sheet, which is stable at all k when $Q = c\kappa/\pi G\sigma \ge 1$ (A. Toomre 1964; J. Binney & S. Tremaine 1987). At the grid scale the lattice departs from the continuum in two ways (Fig. 3b). Self-gravity peaks at $X(\pi) = \pi^2 G\sigma/\Delta$, half the continuum value $2\pi G\sigma k$ at $k = \pi/\Delta$. And pressure vanishes there: in the zigzag, streamlines $j \pm 1$ move together, so $r_{j+1} - r_{j-1}$ and hence σ_j and p_j do not change. The viscous fluxes of J. M. Hahn & J. N. Spitale (2013), which are differenced in the same way, likewise exert no force on the zigzag.

A cold continuum disk ($c = 0$) is unstable to all wavelengths shorter than $\lambda_{\rm crit} = 4\pi^2 G\sigma/\kappa^2$, and the lattice hides these: by Eqn. (C4) a cold lattice is axisymmetrically stable as long as

$$\mathcal{R} \equiv \frac{\lambda_{\rm crit}}{2\Delta} = \frac{2\pi^2 G\sigma}{\kappa^2\Delta} \simeq \frac{\pi G m_r N_r}{\kappa^2 a\Delta a^2} < 2, \tag{C7}$$

where 2Δ is the shortest resolved wavelength, and the last form is for a ringlet of mass m_r, semimajor axis a and width Δa, with $\sigma \simeq m_r/2\pi a\Delta a$ and $\Delta \simeq \Delta a/N_r$. Adding streamlines to a given ringlet raises $\mathcal{R}$ in proportion to N_r.

A viscous ringlet must also avoid the viscous overstability, the growth of axisymmetric epicyclic oscillations that viscous stresses drive when the shear stress rises steeply enough with surface density (U. Schmit & W. M. Tscharnuter 1995; P.-Y. Longaretti & N. Rappaport 1995; H. Salo et al. 2001; J. Schmidt et al. 2001; H. N. Latter & G. I. Ogilvie 2006, 2008). For a cold ringlet with constant kinematic shear and bulk viscosities ν_s and ν_b, as in epi_int_lite, and Keplerian $\kappa = \Omega$, a mode with oscillation frequency $\omega \simeq \kappa(1-x)^{1/2}$ grows to first order in the viscosities at the rate

$$\gamma = \frac{K}{6(1-x)}\left[\nu_s(4x+2) - 3\nu_b(1-x)\right], \tag{C8}$$

where $K = k^2$ and $x = 2\pi G\sigma k/\kappa^2$ in the continuum. Without self-gravity ($x = 0$) the ringlet is stable when $\nu_b > 2\nu_s/3$, the onset found by U. Schmit & W. M. Tscharnuter (1995) and M. Lehmann et al. (2017) for a viscosity independent of σ, while self-gravity is destabilizing (M. Lehmann et al. 2017): a mode is stable only for $x < (3\nu_b - 2\nu_s)/(4\nu_s + 3\nu_b)$, which is 1/7 for $\nu_b = \nu_s$ and 0.4 for $\nu_b = 2\nu_s$. The streamline lattice follows Eqn. (C8) with the grid's versions of both terms: $K = \sin^2\phi/\Delta^2$, from the code's central differences of the viscous stresses, and $x = X(\phi)/\kappa^2$. Both stresses vanish at the zigzag, $\phi = \pi$, where self-gravity is strongest, $X(\pi)/\kappa^2 = \mathcal{R}/2$; so a viscous lattice is stable at every wavelength when $\mathcal{R} < 2/7$ for $\nu_b = \nu_s$, or $\mathcal{R} < 4/5$ for $\nu_b = 2\nu_s$, more stringent than Eqn. (C7). Since $K\nu \sim \nu/\Delta^2$, an overstable lattice's modes grow faster the more streamlines it has.

That limit matters because a ringlet's self-gravity must be strong enough to resist the differential precession due to J_2. Across a ringlet of width Δa, J_2 drives a differential precession $\propto J_2(R_p/a)^2\Omega_a\Delta a/a$, while for a small nonlinearity parameter the ringlet's self-gravity drives one $\propto (m_r/M_p)\Omega_a\Delta\tilde{\omega}(a/\Delta a)^2$, where $\Delta\tilde{\omega}$ is the periapse twist across it (cf. Eqn. C1 of J. M. Hahn et al. 2025). So keeping J_2 from winding up the ringlet requires $m_r \propto J_2\Delta a^3/\Delta\tilde{\omega}$, and then $\mathcal{R} \propto J_2\Delta a N_r/\Delta\tilde{\omega}$: a wider ringlet does not help, and for a given planet more streamlines can be afforded only by a narrower ringlet. This is tested here with simulations of a ringlet of width $\Delta a = 0.0125a$ whose edges are tapered, mass $m_r = 3\times10^{-8}M_p$, eccentricity $e = 10^{-4}$, periapses twisted by π from its inner to outer edge, and kinematic viscosity $\nu = 7\times10^{-11}a^2\Omega_a$, with no pressure, $R_p = 0.5a$ and $J_2 = 0.01$. With $N_r = 100$ streamlines ($\mathcal{R} \simeq 0.09$ in the ringlet's middle, where the taper makes σ 1.25 times and Δ 0.8 times their ringlet-wide means) the ringlet stays smooth for 10^4 orbits while self-gravity, rather than J_2, controls its shape: within about 1000 orbits it straightens the initial twist, and the ringlet's body then stays nearly aligned, as two sub-ringlets that precess slowly relative to each other. With $N_r = 300$ ($\mathcal{R} \simeq 0.3$) and $N_r = 1000$ ($\mathcal{R} \simeq 0.9$) it becomes ragged and its streamlines cross after about 190 and 30 orbits. At $N_r = 1000$ the disturbance starts in the ringlet's interior, grows at the same rate with 40 or 241 particles per streamline, and does not grow when ring gravity is turned off, consistent with the self-gravity-assisted overstability for $\nu_b = \nu_s$ at $\mathcal{R} > 2/7$. Pressure with $Q \leq 2$ delays the crossing by less than two orbits, as expected from Fig. 3b, where $\omega_p^2 \leq 0.1\kappa^2$.

The crossings near $\mathcal{R} \simeq 0.3$ are mostly a different, numerical, problem. The same ringlet without the initial twist, at $N_r = 333$ ($\mathcal{R} \simeq 0.31$), crosses after 145 orbits, and its code's own linearized axisymmetric modes, computed from the reference implementation's accelerations, grow by only 3.3×10^{-3} per orbit, far too slowly. The disturbance that crosses is instead a sawtooth along each streamline with azimuthal wavenumber $m \simeq N_t/2$, where N_t is the number of particles per streamline. It also grows in a circular ringlet, without ring gravity, and without viscosity, though more slowly (after about 220, 170 and 850 orbits), its onset varies non-monotonically with N_t (with $N_t = 30$ or 50 the ringlet survives 10^4 orbits, with 20, 40 or 320 it does not), and pressure with $Q \leq 10$ delays it by at most 20%. It is presumably seeded by the 3-point Lagrange interpolation in longitude that couples neighboring streamlines' particles, and it remains an open problem.

So Eqn. (C7) is a necessary condition only. In practice a simulated ringlet whose self-gravity matters should be run with $\mathcal{R} \lesssim 0.1$, viscous or not, and a ringlet that needs more streamlines must be made narrower and much less massive.

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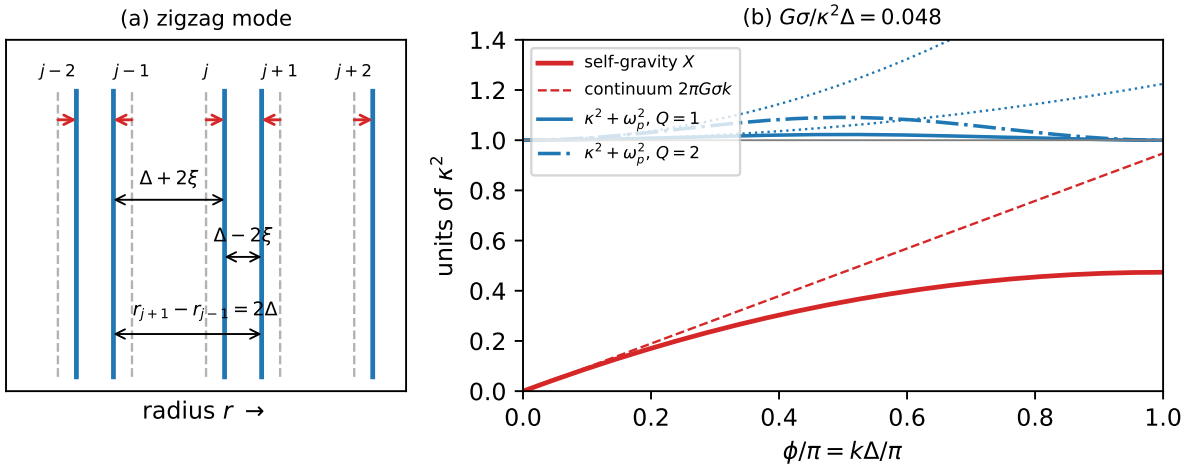


Figure 3. (a) The zigzag mode of a lattice of streamlines (blue) displaced by $\pm\xi$ from their unperturbed radii (dashed): the gaps alternate between $\Delta \pm 2\xi$, but $r_{j+1} - r_{j-1} = 2\Delta$ does not change, so neither does the surface density that sets streamline j's pressure. (b) The terms of Eqn. (C6) versus $\phi = k\Delta$, in units of κ^2, for $G\sigma/\kappa^2\Delta = 0.048$ (the ringlet of the text with $N_r = 1000$): the destabilizing self-gravity X (Eqn. C4, solid red) and its continuum value $2\pi G\sigma k$ (dashed red), and the restoring $\kappa^2 + \omega_p^2$ for $Q = 1$ and 2 (blue; Eqn. C5) and their continuum values $\kappa^2 + c^2k^2$ (dotted blue). A mode is unstable where X exceeds $\kappa^2 + \omega_p^2$.

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